Why Does the Channel Matter?

Separating Attention and Belief Mechanisms in Parental Information Interventions

Jordi Torres Vallverdú

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Information Problems in Education

Students and parents make decisions under imperfect information

- Students hold biased beliefs about returns to education and own ability
  - Major choice and labor market outcomes [Arcidiacono et al., 2024, Wiswall and Zafar, 2015]
  - Persistence and dropout decisions [Stinebrickner and Stinebrickner, 2012]
  - College application and matching [Larroucau and Rios, 2024]
- **Parents are also misinformed** about their children's performance
  - Mean belief–performance gap $>$ 1 SD in some contexts [Dizon-Ross, 2019]
  - Biases concentrated among low-SES households
- Parental investments respond to information $\Rightarrow$ misperceptions distort human capital accumulation

Information Treatments to Parents

Providing information to parents improves outcomes

- Dizon-Ross [2019]: Correcting beliefs about child ability in Malawi
  - Parents reallocate investments toward lower-performing children
  - Beliefs are systematically upward-biased
- Bergman [2021]: SMS-based grade notifications in the US
  - Parents receive same info via SMS vs. standard report cards
  - Result: 0.19 SD GPA increase at very low cost
  - $\sim$59% from belief revision, $\sim$41% from reduced monitoring costs
- **The puzzle:** the *channel* of delivery matters, not just content

Why Does the Channel Matter? Possible Mechanisms

Standard economics: only content should matter $\Rightarrow$ **three behavioral mechanisms**

1. **Biased beliefs & asymmetric updating**
   - Parents hold upward-biased priors; update asymmetrically
   - More information $\Rightarrow$ belief correction
2. **Limited attention & salience** $\leftarrow$ *main focus*
   - Under cognitive load, "child performance" drops out of active decision model [Gabaix, 2014]
   - SMS creates high salience contrast vs. report card among paperwork [Bordalo et al., 2012]
3. **Present bias & procrastination**
   - Parents intend to engage but delay; SMS reduces action delay

Problem: existing designs bundle content and channel $\Rightarrow$ cannot separate mechanisms

Research Questions

Goal: Disentangle the mechanisms behind parental information treatments

1. Does changing the *channel* alone (holding content fixed) affect parental behavior?
   $\Rightarrow$ Attention / salience mechanism

2. Does providing richer *content* (holding channel fixed) affect behavior?
   $\Rightarrow$ Belief correction mechanism

3. Are channel and content *complements* or *substitutes*?
   $\Rightarrow$ Interaction between attention and beliefs

4. Does *frequency* of contact matter beyond content and channel?
   $\Rightarrow$ Sustained attention vs. one-shot correction

Model: Production Function & Belief Updating

Grade production: Student i in school e:

$$Y_i = Y(a_i,\, e_i^p,\, e^s) + \varepsilon_i$$

- a_i: student ability $\quad e_i^p$: parental effort $\quad e^s$: school effort

Belief updating: Parents hold prior $\hat{a}_i^{\text{pre}}$, receive signal σ_i:

$$\hat{a}_i^{\text{post}} = \hat{a}_i^{\text{pre}} + \underbrace{A(\kappa_i)}_{\text{attention}} \cdot \underbrace{\lambda(\sigma_i, \hat{a}_i^{\text{pre}})}_{\text{salience}} \cdot (\sigma_i - \hat{a}_i^{\text{pre}})$$

- $A(\kappa_i) \in [0, 1]$: attention weight [Gabaix, 2014]
  - High cognitive load $\kappa_i \Rightarrow$ low attention; SMS lowers κ_i
- $\lambda(\cdot) \in [0, 1]$: salience weight [Bordalo et al., 2012]
  - Signals contrasting with priors get more weight

Model: Effort Choice

Parental effort: Given updated beliefs, parent maximizes:

$$\max_{e_i^p \geq 0} \ln Y(a_i,\, e_i^p,\, e^s)\Big|_{\hat{a}_i^{\text{post}}} - C^p(e_i^p,\, z_i)$$

- $C^p(e_i^p, z_i)$: cost of effort, heterogeneous in parent type z_i (SES, time)
- Allows credit constraints to enter through cost function

Key distinction:

- $A(\kappa_i)$: determines whether parent *considers* the information at all
- $\lambda(\cdot)$: determines how much weight the signal gets *conditional on attention*

$\Rightarrow$ Both needed: a parent can attend to a signal yet underweight it (or vice versa)

The RCT: Five Treatment Arms

Setting: Primary/secondary schools with digital gradebooks but low parent–school communication. Randomization stratified by school characteristics.

Arm	Description	Channel	Content
C	Standard report cards (status quo)	Paper	Standard
T1	Same info, delivered via SMS/WhatsApp	SMS	Standard
T2	Detailed info (grades, ranking, absences)	Paper	Detailed
T3	Detailed info via SMS (replicates Bergman)	SMS	Detailed
T4	Same as T3, double frequency (weekly)	SMS	Detailed+

- T1 vs. C: isolates *channel* (attention)
- T2 vs. C: isolates *content* (beliefs)
- T3 vs. C: joint effect (replication)
- T4 vs. T3: frequency / sustained attention

Identification Strategy

2×2 **factorial design** enables clean decomposition:

	Standard content	**Detailed content**
Paper channel	C	T2
SMS channel	T1	T3

Estimating equation:

$$Y_i = \alpha + \beta_1 \cdot \mathbf{1}[\text{SMS}]_i + \beta_2 \cdot \mathbf{1}[\text{Detailed}]_i + \beta_3 \cdot \mathbf{1}[\text{SMS} \times \text{Detailed}]_i + X_i'\gamma + \epsilon_i$$

- β_1: attention effect A (channel changes, content fixed)
- β_2: belief-correction effect through $\lambda \cdot (\sigma - \hat{a})$
- β_3: complementarity between attention and beliefs
- T4 arm: adds frequency dimension

Additional data: Elicit baseline beliefs $\hat{a}_i^{\text{pre}}$ to observe belief gap $\Rightarrow$ helps separate biased priors from distorted updating

Power Calculations

Benchmark: Bergman [2021] finds 0.19 SD effect (joint treatment)

Assumptions:

- MDE $= 0.10$ SD (individual mechanisms are smaller than joint)
- Significance level $\alpha = 0.05$, power $= 0.80$
- ICC $\rho = 0.05$ (school-level clustering)
- 20 students per school cluster

Required sample (per arm):

$$n = \frac{(z_{1-\alpha/2} + z_{\text{power}})^2 \cdot 2\sigma^2 \cdot \text{DEFF}}{\text{MDE}^2} \approx \frac{(1.96 + 0.84)^2 \cdot 2 \cdot 1.95}{0.01} \approx 3{,}060$$

where DEFF $= 1 + (20 - 1) \times 0.05 = 1.95$

- $\approx$ 3,000 students per arm $\times$ 5 arms $=$ **15,000 students total**
- $\approx$ 750 school clusters (150 per arm)
- Stratification by school SES helps reduce variance

Conclusion & Policy Implications

Summary:

- Existing evidence shows information treatments improve educational outcomes
- But we cannot distinguish *why*: attention, beliefs, or their interaction
- Proposed RCT with factorial design separates these mechanisms

Policy implications depend on the mechanism:

- If $\beta_1 \gg \beta_2$: channel/salience drives the effect $\Rightarrow$ invest in **nudge design**
- If $\beta_2 \gg \beta_1$: content/beliefs drive the effect $\Rightarrow$ **one-shot information** suffices
- If $\beta_3 > 0$: attention and beliefs are complements $\Rightarrow$ **both needed together**

Contribution:

- Moves beyond reduced-form to structural understanding of why information works
- Low-cost interventions with potentially large returns
- Framework generalizable to other parental/educational information settings

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